ELECENG 2EI4

Bonus Project: 3-bit DAC

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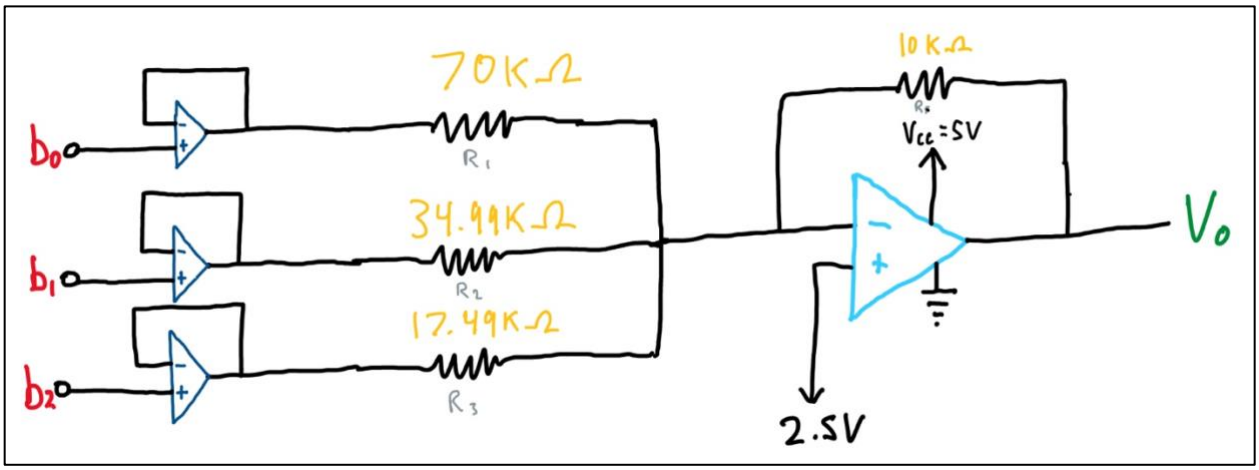
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Summary

The project requires the design, simulation, and building of a 3-bit digital-to-analog converter with full-scale analog voltage, $V_{FS} = 5V$. This was to be done using components in the 2CI4 and 2EI4 kits, as well as DC supplies from the analog discovery. The solution is an inverting weighted-summer op-amp, which in hardware is in the LMC662 chip. The feedback resistor was 10kΩ, and the three inputs into the inverting input had a resistor weight ratio of 4:2:1. To ensure bit line current was nearly 0, each line passed through a voltage buffer before the resistor and the main op-amp. The non-inverting input was set to 2.5V to shift the output into the 0-5V range. The solution in practice displays good transfer characteristics, by reaching 0.038V and 4.97V, for near rail-to-rail performance. The step sizes in practice were very close to uniform, as the maximum differential non-linearity was only 0.048 LSB. In addition, the offset error was only 0.038 V, and the gain error was only -0.068V, meaning the solution overall performs well for a real-life model.

Design

i. Circuit Diagram



ii. Functionality

Three digital inputs, each being one of the three bits pass through distinct resistances on their way to the summing node (inverting input). These resistances are chosen so the most significant bit has the largest effect on the output, the least significant bit has the lowest effect, and the middle bit has a medium effect. For each input, it first passes through a unity gain buffer, as its 1TΩ input resistance draws very little current at even 5V. The buffer being unity gain means it can send an equivalent

voltage to the binary-weighted resistor as if it wasn't even there. The non-inverting input is biased at 2.5V due to the 0-5V supplies. This means the inverting input is at the same reference level, due to the negative feedback. The output voltage adjusts so KCL at the inverting input is satisfied, producing an analog voltage. The three binary bits are driven by either 0V for logic high, or 5V for logic low, meaning a sequence 5V, 0V, 0V, would be 011, or 3. In the circuit, when a bit is high, that branch's effect is sent to the input, and when it is low, it sends the opposite state, thus producing eight distinct voltages for the $2^3 = 8$ combinations.

iii. Calculations

For a weighted-summer, each input's contribution to the output is determined by the ratio $\frac{R_f}{R_i}$. From there, the ratios are chosen so the bits represent their binary value.

Given the DAC is 3-bit, $V_{FS} = 5V$, the desired output has 8 equal levels from 0 to 5V, and the weightings of b_2, b_1, b_0 are 4, 2, 1:

$$LSB = \frac{V_{FS}}{2^3 - 1} = \frac{5}{7}$$
$$D = 4b_2 + 2b_1 + b_0$$

$$V_o = \frac{5}{7}(4b_2 + 2b_1 + b_0)$$

$$MSB\ contribution: \frac{20}{7}$$
$$Middle\ bit\ contribution: \frac{10}{7}$$
$$LSB\ contribution: \frac{5}{7}$$

Each branch factors into the output proportionally to $\frac{R_f}{R_i}$, but output differs by 5V between states, so each contribution is $5(\frac{R_f}{R_i})$

$5(\frac{R_f}{R_{b2}}) = \frac{20}{7}$	$5(\frac{R_f}{R_{b1}}) = \frac{10}{7}$	$5(\frac{R_f}{R_{b0}}) = \frac{5}{7}$
$\frac{R_f}{R_{b2}} = \frac{4}{7}$	$\frac{R_f}{R_{b1}} = \frac{2}{7}$	$\frac{R_f}{R_{b0}} = \frac{1}{7}$

Feedback resistor chosen:

$$R_f = 10k\Omega$$

$\frac{10k}{R_{b2}} = \frac{4}{7}$ $70k = 4R_{b2}$ $17.5k = R_{b2}$	$\frac{10k}{R_{b1}} = \frac{2}{7}$ $70k = 2R_{b1}$ $35k = R_{b1}$	$\frac{10k}{R_{b0}} = \frac{1}{7}$ $70k = R_{b0}$

Voltage divider with equivalent resistors used between 0 and 5V for 2.5V bias:

$$V_+ = 5\left(\frac{R}{R+R}\right) = 2.5V$$

Using an op-amp as a unity gain buffer for the bit lines, keeps current near-zero, as the only input resistance on this line will be that of the LMC662 chip, which is incredibly large at 1TΩ.

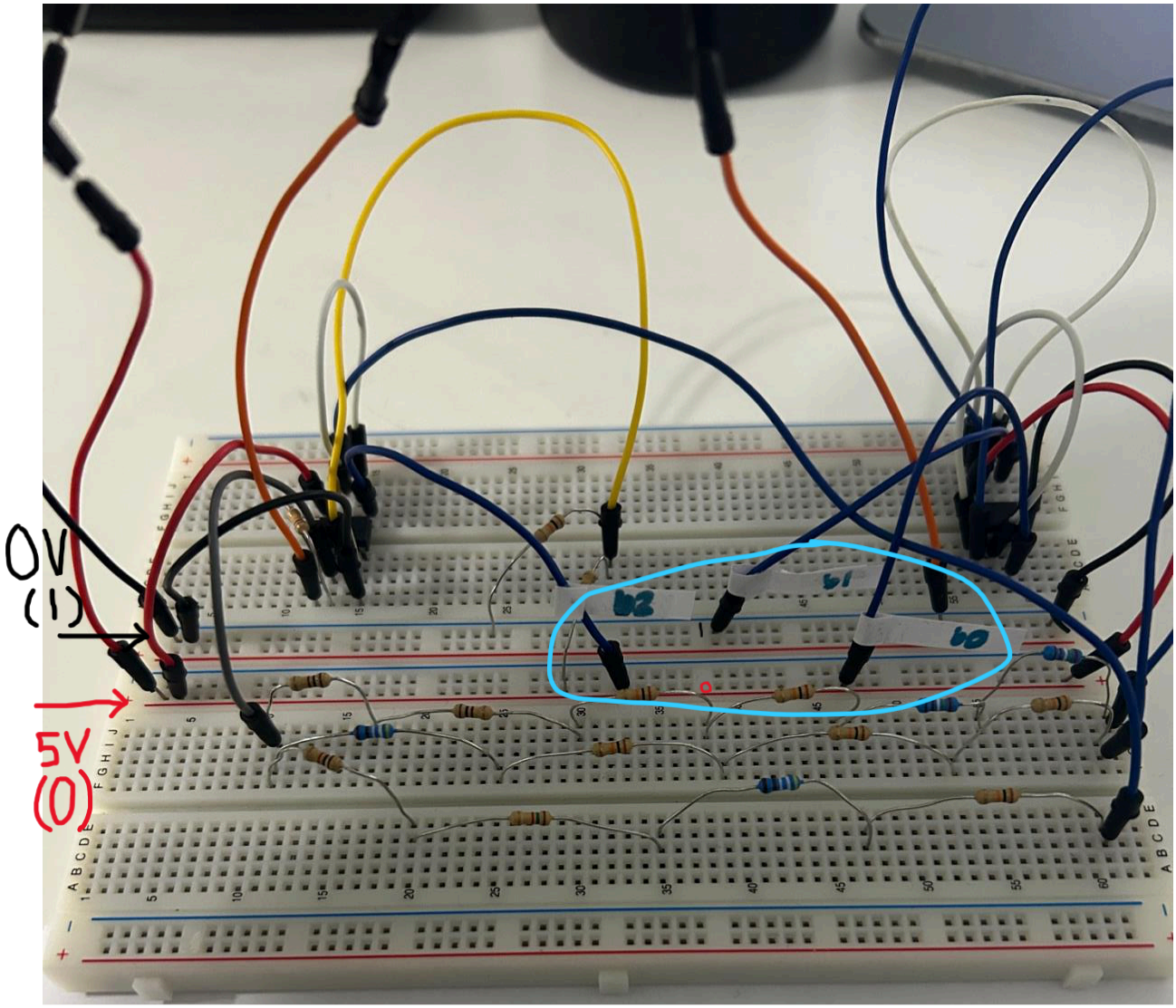
$$I_{max} = \frac{V_{max}}{R_{in}}$$
$$I_{max} = \frac{5}{1\text{T}} = 5pA$$

iv. Circuit reference

The circuit design used came from Lecture 2: “op-amp applications” from the COMPENG 2CF3 Circuits and Waves course. The lecture defines the circuit as an inverting summer and describes the dependency on the $\frac{R_f}{R_i}$ ratio. The resistor weighting, unity gain buffers, and 2.5V bias were the adaptations made for this specific application.

Measurement and Analysis

i. Physical circuit



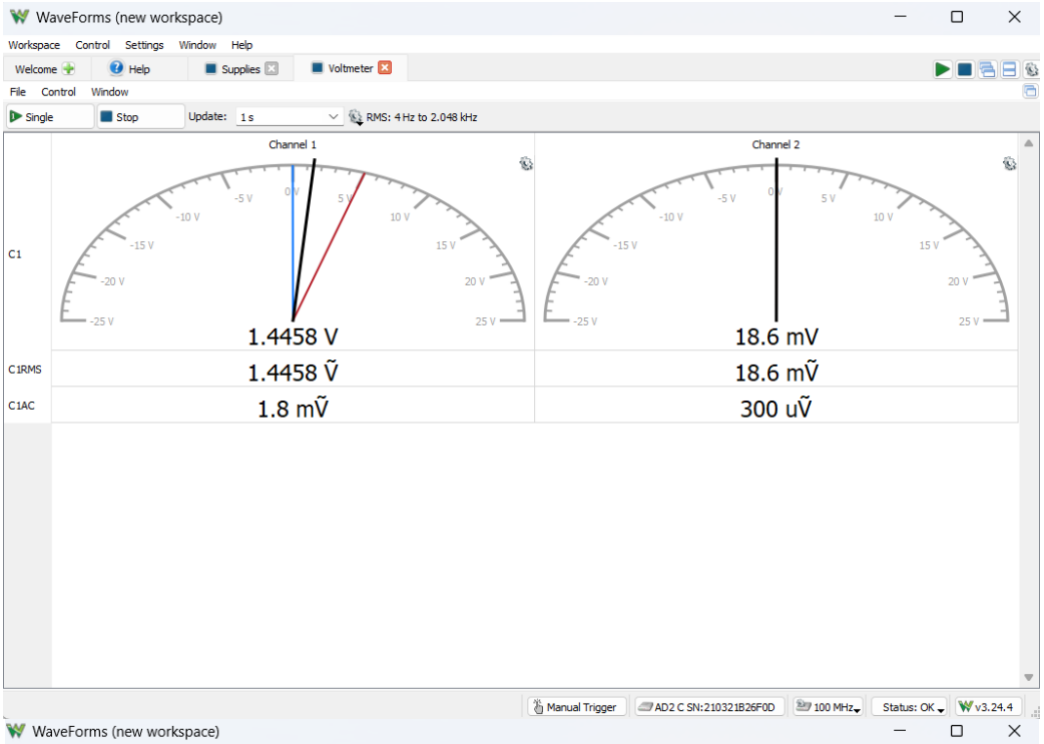
ii. Digital inputs

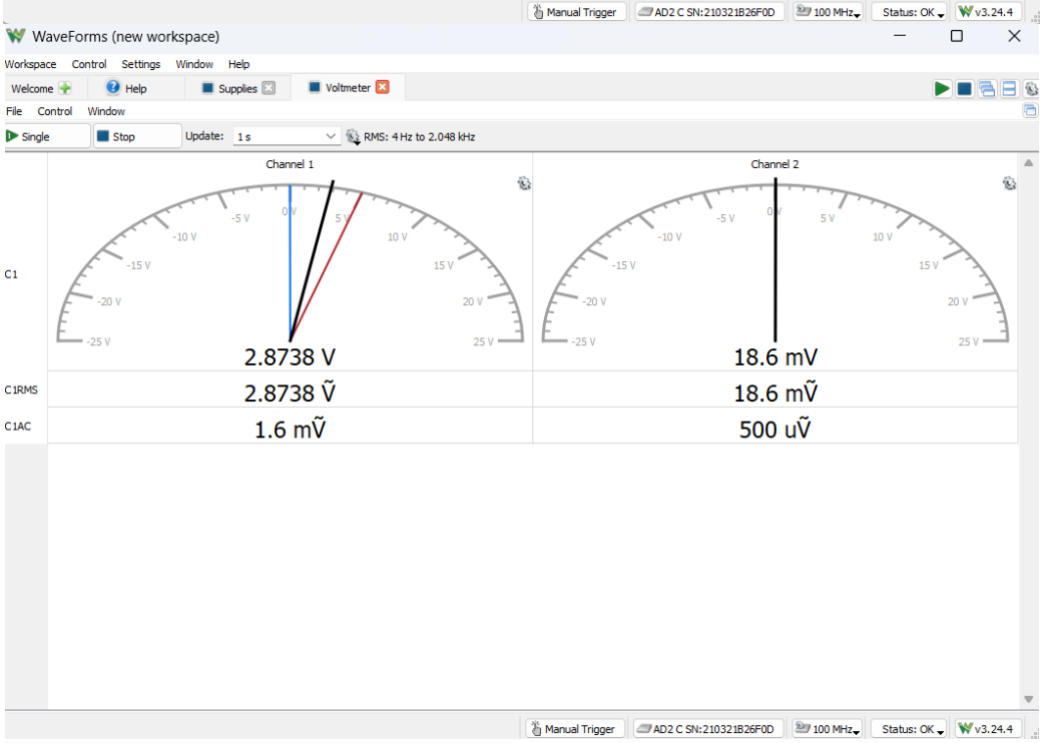
5V DC was driven from the analog discovery to a rail for logic low (0), and ground from the analog discovery was connected to a different rail for logic high (1), as seen in the circuit picture. Before each branch went through its buffer and resistor, it was fed by a jumper that could be connected to either rail to make its value either 0 or 1, for a visually intuitive setup.

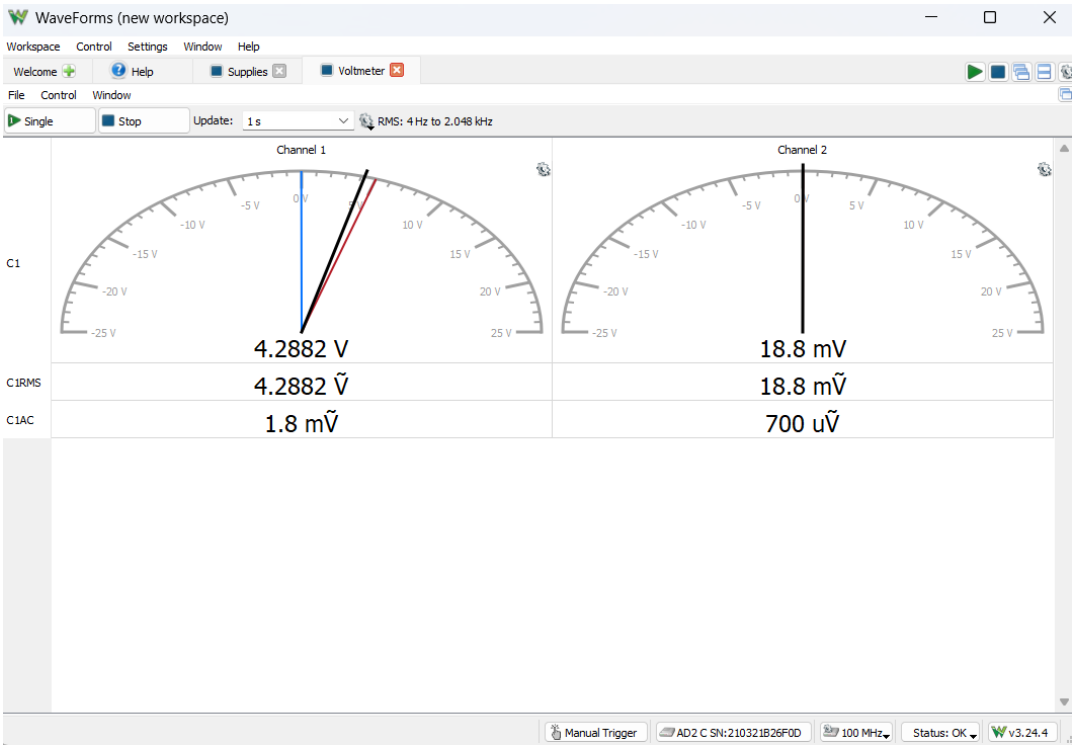
iii. Transfer Characteristics

All 8 possible input combinations were tested, and voltage was measured at the output for each. The measured values match the ideal values very closely, meaning the design functions correctly. The table below summarizes these results, and the screenshots below it display the voltmeter for 2, 4, and 6, for example.

Voltages (b_2, b_1, b_0)	**Binary Value**	**Decimal Value**	**Ideal Output voltage (V) $= i \times (5/7)$**	**Measured Output Voltage (V)**
555	000	0	0	0.038
550	001	1	0.714	0.732
505	010	2	1.429	1.45
500	011	3	2.143	2.16
055	100	4	2.857	2.87
050	101	5	3.571	3.58
005	110	6	4.286	4.29
000	111	7	5	4.97


WaveForms (new workspace)
Workspace Control Settings Window Help
Welcome Help Supplies Voltmeter
File Control Window
Single Stop Update: 1 s RMS: 4 Hz to 2.048 kHz
Channel 1
Channel 2
C1
1.4458 V
18.6 mV
C1RMS
1.4458 Ṽ
18.6 mṼ
C1AC
1.8 mṼ
300 uṼ
Manual Trigger
AD2 C SN:210321B26F0D
100 MHz
Status: OK
v3.24.4


WaveForms (new workspace)
Workspace Control Settings Window Help
Welcome Help Supplies Voltmeter
File Control Window
Single Stop Update: 1 s RMS: 4 Hz to 2.048 kHz
Channel 1
Channel 2
C1
2.8738 V
18.6 mV
C1RMS
2.8738 Ṽ
18.6 mṼ
C1AC
1.6 mṼ
500 uṼ
Manual Trigger
AD2 C SN:210321B26F0D
100 MHz
Status: OK
v3.24.4



iv. Gain Error

The gain error is a measure of the spanning accuracy of the output voltage. It verifies that from the lowest value to the highest value, the output voltage differs by the full-scale voltage. It was determined by calculating the span of the measured output voltage and finding the difference from the full-scale voltage.

$$span = V_o(7) - V_o(0)$$
$$span = 4.97 - 0.038 = 4.932$$

$$E_{gain} = 4.932 - 5 = -0.068$$

This value means the span of values measured is only 0.068V less than the full-scale voltage, meaning this design spans the full-scale voltage fairly well.

v. Maximum Differential Non-Linearity

Differential non-linearity refers to the accuracy of each step size of output voltages between values. Ideally, this value should be the LSB, which is 0.7143 (5/7). To measure it, each voltage step was calculated using $V_o(i) - V_o(i-1)$. Each of these values were then ratioed with the ideal LSB and 1 was taken away, using $E_{DNL} = \frac{\Delta V}{0.7143} - 1$. Of all steps, the one from 6 to 7 had the highest magnitude of 0.048 LSB,

which means the maximum deviation from the ideal voltage step for this design is only 4.8% of a LSB, so it generally has good linearity.

vi. Offset Error

The offset error is simply the shift in the output voltage at the initial value, which ultimately shifts all values. Ideally it would be 0. This value is just the first measurement, $V_o(0)$, which for this design was 0.038V, meaning the output voltage started very slightly above 0, which is fair.

Discussion

i. Error Origin

Despite the errors measured being quite small, they are still errors. However, this does not mean the design is flawed. The first source of error is the LMC662 chip used. This chip does not produce perfect rail-to-rail performance in practice, which likely contributed to the offset and gain errors. The chip has a slight voltage difference between its inputs, which is another factor in the offset error. In addition, for simplicity purposes, 17.49kΩ and 34.99kΩ had to be used, which could slightly change the resistor ratios and binary weights, affecting the gain error and maximum differential non-linearity error. Finally, breadboard debris, imperfect wiring, internal resistances, and manual input switching, which are all normal in real-world practice, will inevitably deviate measured results from theoretical ones for all errors.

ii. Alternate Designs

Other implementations were not strongly considered due to familiarity with the structure and calculations for the inverting summer from COMPENG 2CF3. In addition, it led to resistor values compatible with the kits at hand. One other alternative design however is the R-2R ladder DAC. This implementation only uses two resistors, even for 3-bits, which in real-world applications would reduce costs and increase scalability. One downside to this implementation is that it would require very precise resistor matching for 3 bits, which could have created problems given the resistors available.

References

[1] A. S. Sedra, K. C. Smith, T. C. Carusone, and V. Gaudet, *Microelectronic Circuits*, 8th ed. New York, NY, USA: Oxford University Press, 2020.

[2] “Lecture 2: Operational Amplifiers - Applications,” COMPENG 2CF3 Circuits and Waves lecture notes, uploaded PDF.

[3] Texas Instruments, LMC660, LMC662 CMOS Dual Operational Amplifiers, rev. SNOSC51D, Feb. 2024.